Monthly Rainfall Analysis and Regression Pipeline

Mathematical Implementation and Usage

Rainfall-only component for Sringeri and Sulya, Karnataka

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Abstract

This document describes the implemented rainfall-only pipeline for the supplied Sringeri and Sulya monthly workbooks, and the separate local-file ingestion path prepared for official IMD rainfall. It covers missing-data handling, calendar-aligned features, Random Forest and radial-basis Support Vector Regression (SVR), time-ordered validation, and source-aware IMD ingestion. No IMD file was downloaded during this run; the document distinguishes source metadata from data actually ingested. The project does not detect invasive species or process imagery. The workbook headings omit the unit, but the official Karnataka WRIS Rainfall Analytics page provides the matching WRD-SMS

source selection and labels the rainfall measures in mm. Values are treated as mm without conversion. The WRIS page marks data as provisional and subject to revision (https://kwris.aciwrm.org/RainfallAnalytics.aspx, accessed 2026-10-20).

1 Scope and data

Each workbook contains a sheet named `RainfallConsolidatedReport`. The data table is in wide form: one row per year, with twelve columns for January through December. The loader finds the `Year` row and month columns, then converts each year-month pair into a monthly time stamp. A monthly observation is the recorded value in that workbook cell.

The observed series span January 2021 through October 2026, with 61 recorded values and 11 missing calendar months per location. April–December 2023 and November–December 2026 are blank in both workbooks. Table coverage extends through December 2026 because those month columns exist; October 2026 is the final observed month. The unit is confirmed from the matching WRD-SMS source on Karnataka WRIS, while the completeness convention and reporting cut-off remain undocumented. This provenance and the provisional-data caveat are recorded in the quality report and model metadata.

Let $y_{\ell,t}$ be the monthly rainfall number for location ℓ and month t, where $\ell \in \{\text{Sringeri}, \text{Sulya}\}$. A blank workbook cell is represented by a missing value, denoted NA, rather than by zero. A recorded zero remains 0. No target values are interpolated or imputed. Negative numeric rainfall values are invalid and cause the pipeline to stop for review. Repeated location-month records are collapsed only if their nonmissing values agree; conflicting repeats stop processing.

The monthly calendar is reindexed independently for each location. This ensures that a six-month lag means six calendar months and does not mean the sixth previous nonmissing row. All operations preserve the location boundary.

2 Separate official IMD ingestion path

The supplied WRD-SMS Excel workbooks remain the primary modelling inputs. IMD data are inspected and processed by a separate command, and the ingestion code does not append them to the existing WRD-SMS modelling table. This separation is important because source, measurement method, temporal scale, and spatial support are not interchangeable.

2.1 Sources and acquisition status

The IMD Pune page describes a daily $0.25° \times 0.25°$ grid for 1901–2024, with 135×129 cells, rainfall in mm, and -999 as the undefined-value marker. Its sample reader documents coordinate origins and increments. The page does not document the downloaded binary byte order, so the local grid reader requires the user to verify and specify byte order; it checks the expected file size and plausible selected-cell values before returning a series. The source page and citation guidance are at https://imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html.

The National Water Data Portal (NWDP) page describes its IMD data as observed rainfall measurements and lists district-wise and state-wise daily CSV resources: https://nwdp.nwic.gov.in/dataset/rainfall-daily-imd. In this project run, the CSV requests could not be retrieved: the web fetcher rejected the CSV content type (and one direct request was inaccessible), while shell network access failed DNS resolution. No NWDP file was downloaded. Consequently, its column schema, row-level spatial support, unit, date coverage, and missing-value markers are explicitly pending local inspection; no NWDP coverage or rainfall result is reported.

The complete acquisition log and manual steps are in `docs/imd_data_acquisition_report.md`. Original downloads should be saved unchanged under `data/official/imd/incoming/`. First inspect a CSV; then review actual headers, metadata, identifiers, date convention, units, and

missing-value convention before ingestion. The source inventory is `outputs/imd/source_catalogue.csv`; field definitions and rules are in `outputs/imd/data_dictionary.md`.

2.2 Source types, units, and missing data

The normalized IMD schema labels each row with a source type (station, district, state, or gridded). The original numeric value and verified source unit are retained alongside canonical mm. For an explicitly verified source unit u, canonicalization is

$$r_{\mathrm{mm}} = \begin{cases} r_u, & u = \mathrm{mm}, \\ 10 r_u, & u = \mathrm{cm}. \end{cases} \tag{1}$$

The NWDP CSV command requires an explicit unit argument and rejects unsupported/unknown units. This argument must be based on the local source documentation, not guessed from the numeric range or a column name. Blank and configured missing values are represented as missing and flagged; they are not replaced with zero. The grid's documented -999 marker is translated to missing. Duplicate source keys (station or region and date, where present) stop ingestion for review rather than being silently removed.

2.3 Spatial assignment and comparison

Station identifiers and coordinates are retained as supplied and are not mapped automatically to Sringeri or Sulya. A station comparison requires a selected station ID, constant source coordinates, and explicit operator attestation that the station ID and coordinates were verified against authoritative metadata. District-level or state-level rows are never assigned to a taluk. The spatial row schema of the NWDP district-wise file remains unverified until the actual CSV is obtained and inspected.

For IMD gridded data, the user supplies verified site coordinates (ϕ, λ). The implemented selection is the nearest grid-cell center, with no interpolation:

$$i^* = \arg\min_i |\lambda_i - \lambda|, \qquad j^* = \arg\min_j |\phi_j - \phi|, \tag{2}$$

where grid centers begin at 66.5° E and 6.5° N and are spaced by 0.25°. The selected center and coordinate offsets are recorded. A grid value represents an area and is not a rain-gauge observation at the taluk center or a verified station. Comparisons with WRD-SMS are opt-in and only use matched, complete monthly totals. Even then, the plot labels their differing spatial supports; it is a source comparison, not a claim that the measurements are equivalent.

Daily values are aggregated to monthly totals by summing observed daily rainfall. The output records observed-day count, expected days in that calendar month, and a completeness flag. Only complete IMD months are eligible for source-comparison plots. Individual station series remain separated; the code never sums rainfall across stations to create a single series.

2.4 IMD ingestion commands

With the virtual environment active, use the commands below. The inspection command records headers, a small row sample, file size, delimiter, and SHA-256 without interpreting the data. Replace placeholder column names and unit only after reviewing the downloaded source file and metadata.

```
python scripts/ingest_imd_rainfall.py inspect-csv \
  data/official/imd/incoming/rainfall_districtwise_daily_imd.csv

python scripts/ingest_imd_rainfall.py ingest-csv \
  data/official/imd/incoming/rainfall_districtwise_daily_imd.csv \
```

```
  --source-type imd_district \
  --date-column '<verified date header>' \
  --rainfall-column '<verified rainfall header>' \
  --location-column '<verified region header>' \
  --unit '<verified mm or cm>' \
  --missing-marker '<source-documented sentinel>'
```

If the reviewed NWDP file contains station-level rows, classify it as station data and verify the station ID and coordinates before requesting a comparison:

```
python scripts/ingest_imd_rainfall.py ingest-csv \
  data/official/imd/incoming/verified_station_rainfall.csv \
  --source-type imd_station \
  --date-column '<verified date header>' \
  --rainfall-column '<verified rainfall header>' \
  --station-id-column '<verified station ID header>' \
  --latitude-column '<verified latitude header>' \
  --longitude-column '<verified longitude header>' \
  --station-id '<verified station ID>' \
  --unit '<verified unit>' \
  --compare-wrd-location Sringeri --site-match-verified
```

For a local gridded year, supply a verified coordinate and byte order from the actual downloaded file's metadata/readme:

```
python scripts/ingest_imd_rainfall.py ingest-grid-year \
  data/official/imd/incoming/ind2024_rfp25.grd \
  --year 2024 --latitude '<verified latitude>' \
  --longitude '<verified longitude>' \
  --byte-order '<verified little or big>' --label '<location label>'
```

Successful ingestion writes source-separated daily and monthly CSVs, metadata JSON, and historical plots under `outputs/imd/`. A gridded comparison plot is produced only when `-compare-wrd-location Sringeri` or `Sulya` is explicitly supplied. Station comparison additionally requires `-station-id`, `-compare-wrd-location`, and `-site-match-verified`, as well as source station ID and coordinate columns. District/state aggregates cannot be compared as if they were taluk measurements.

3 Feature construction

For each location ℓ and calendar month t, the target is the current monthly observation $y_{\ell,t}$. The implemented predictor vector is

$$\mathbf{x}_{\ell,t} = \left[s_t, c_t, y_{\ell,t-1}, y_{\ell,t-2}, y_{\ell,t-3}, y_{\ell,t-6}, y_{\ell,t-12}, r^{(3)}_{\ell,t}, r^{(6)}_{\ell,t} \right], \tag{3}$$

where the cyclic month features are

$$s_t = \sin\left(\frac{2\pi m_t}{12}\right), \qquad c_t = \cos\left(\frac{2\pi m_t}{12}\right), \qquad m_t \in \{1, \ldots, 12\}. \tag{4}$$

The k-month lag is the value exactly k calendar months earlier. The prior-only rolling mean of width w is

$$r^{(w)}_{\ell,t} = \frac{1}{w} \sum_{j=1}^{w} y_{\ell,t-j}, \qquad w \in \{3, 6\}. \tag{5}$$

It is defined only when all w prior monthly values are observed. In particular, neither rolling feature nor any lag contains $y_{\ell,t}$ or a future value. A missing value in a lag window remains missing; observations are never inserted to make the window complete.

The features are calculated separately for each location and are exported in `outputs/rainfall_features.csv` and `outputs/rainfall_monthly_features.csv`. The latter is a join-friendly table with location, year, month, date, observed rainfall, quality flag, and the valid features. Model predictions are kept in a separate file and never replace observed rainfall.

4 Training-fold preprocessing

Some predictor values are unavailable near the start of a series or after a missing month. Each model pipeline applies median imputation to predictors. For a training fold F, feature j's imputation value is

$$\widetilde{x}_{F,j} = \text{median}\{x_{i,j} : i \in F,\ x_{i,j} \text{ observed}\}. \tag{6}$$

The imputer is fitted on that fold's training rows only, then applied to its validation rows. The target y is never imputed. SVR additionally standardizes predictors using training-fold means and standard deviations:

$$z_{i,j} = \frac{x_{i,j} - \mu_{F,j}}{\sigma_{F,j}}. \tag{7}$$

If a feature has zero training-fold standard deviation, the scaler handles it without division by zero. Random Forest uses the same training-fold median imputation but does not require feature scaling.

5 Regression models

5.1 Random Forest

For each location, the Random Forest is an ensemble of B regression trees. Its prediction is the arithmetic mean of tree predictions:

$$\widehat{y}^{\text{RF}}_{\ell,t} = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}_{\ell,t}). \tag{8}$$

The implementation compares $B \in \{100, 300\}$, maximum depth $\{3, \text{unlimited}\}$, minimum leaf size $\{1, 3\}$, and minimum split size $\{2, 5\}$. A fixed random seed of 42 is used. Candidate settings are ranked by expanding-window cross-validation MAE on the pre-holdout data.

5.2 Radial-basis Support Vector Regression

SVR uses the radial-basis kernel

$$K(\mathbf{z}_i, \mathbf{z}_j) = \exp\bigl(-\gamma\|\mathbf{z}_i - \mathbf{z}_j\|_2^2\bigr) \tag{9}$$

and the ϵ-insensitive loss. Its standard primal optimization can be written as

$$\min_{\mathbf{w},b,\boldsymbol{\xi},\boldsymbol{\xi}^*} \quad \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_i(\xi_i + \xi_i^*) \tag{10}$$

$$\text{subject to} \quad y_i - (\mathbf{w}^\mathsf{T}\phi(\mathbf{z}_i) + b) \le \epsilon + \xi_i, \tag{11}$$

$$(\mathbf{w}^\mathsf{T}\phi(\mathbf{z}_i) + b) - y_i \le \epsilon + \xi_i^*, \tag{12}$$

$$\xi_i, \xi_i^* \ge 0. \tag{13}$$

The limited search compares $C \in \{1, 10\}$, $\epsilon \in \{0.1, 1\}$, and $\gamma \in \{\text{scale}, 0.1\}$. Scaling and imputation are part of an sklearn pipeline, so their parameters are learned inside each training fold.

5.3 Baselines

Two simple one-step-ahead baselines are reported where their required prior observation is present:

$$\widehat{y}_{\ell,t}^{\text{persist}} = y_{\ell,t-1}, \tag{14}$$

$$\widehat{y}_{\ell,t}^{\text{seasonal}} = y_{\ell,t-12}. \tag{15}$$

They provide reference errors; neither is tuned.

6 Chronological validation and selection

No row shuffling is used. For each location, the final 12 calendar months ending at the last observed month form the holdout. In this run, the holdout period is November 2025–October 2026. Only observed targets within that calendar interval are scored. Earlier observations are the training period. Hyperparameters are chosen without using the final holdout.

Model selection uses expanding-window `TimeSeriesSplit` on the pre-holdout observations. In each successive fold, the training prefix grows and the next block is used for validation. The selected setting minimizes the mean fold MAE. Imputers, scalers, and estimators are refitted on each fold's training prefix. The final evaluation estimator is then fit on all pre-holdout training observations and evaluated once on the reserved holdout.

This is a rolling one-month-ahead setup: when predicting month $t+1$, the observed rainfall in month t may be used as a lag if it has become available. Validation rows remain in chronological order. Because TimeSeriesSplit is applied to observed target rows, a 12-row fold does not always span 12 calendar months. For this data, fold 1 spans February 2022–January 2023, fold 2 spans February 2023–October 2024 (the nine-month 2023 target gap is inside this window), and fold 3 spans November 2024–October 2025. Each has 12 observed targets. The final saved models are the pre-holdout evaluation models; the holdout observations are not used to fit them.

7 Evaluation metrics

For observed test values y_i and predictions $\widehat{y}_i$, with n evaluated observations, the metrics are

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \widehat{y}_i|, \tag{16}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \widehat{y}_i)^2}, \tag{17}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \widehat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{18}$$

R^2 is undefined when fewer than two nonconstant observations are available; the output records it as missing in that case. MAE and RMSE use millimetres, the unit used by the matching WRD-SMS view in the Karnataka WRIS source. Cross-validation results and holdout results are stored separately and must not be conflated.

8 Results for the supplied workbooks

The executed pipeline selected Random Forest settings by cross-validation. Its observed holdout metrics are shown below, in mm.

Location	Model	MAE	RMSE	R^2
Sringeri	Random Forest	253.23	344.23	0.563
Sringeri	SVR	282.14	535.53	-0.056
Sringeri	Seasonal naive	258.10	404.90	0.396
Sringeri	Persistence	284.58	531.99	-0.043
Sulya	Random Forest	203.23	287.50	0.441
Sulya	SVR	235.14	367.25	0.088
Sulya	Seasonal naive	233.28	313.72	0.334
Sulya	Persistence	219.92	364.59	0.101

Random Forest has the lowest holdout MAE of the listed methods for both locations in this run. This result is descriptive of one short holdout, not evidence of dependable long-term generalization. For cross-validation, mean fold MAE was 290.88 (Random Forest) and 305.52 (SVR) for Sringeri; it was 160.12 (Random Forest) and 287.93 (SVR) for Sulya. Per-fold rows and all model parameters are in `outputs/cross_validation_results.csv` and `outputs/model_metrics.csv`.

The selected-parameter fold MAEs expose period sensitivity. Sringeri Random Forest has fold MAE 377.09, 245.82, and 249.73; its first fold is harder. Sringeri SVR has 417.24, 103.19, and 396.14, so its mean is substantially helped by the unusually easy middle fold. Sulya Random Forest gives 206.75, 156.09, and 117.53, improving over later folds rather than showing a constant error. Sulya SVR is more similar across folds at 288.33, 265.97, and 309.49, but has higher MAE. Three folds are too few to establish stable expected error.

8.1 Random Forest heavy-rainfall behavior

The actual-versus-predicted plots show underprediction of the single August peak in both locations: 1,588.23 mm observed versus 1,115.47 mm predicted at Sringeri, and 1,181.8 mm versus 780.88 mm at Sulya. The pattern is not a uniform underprediction of all heavy-rainfall months. For example, June is overpredicted by about 390 mm at Sringeri and 405 mm at Sulya; July is underpredicted at Sringeri by about 221 mm but overpredicted at Sulya by about 84 mm. The forest misses the largest peak, while its bias depends on month and location.

The selected Random Forest configuration in the reproducible rerun was 300 trees, depth 3, minimum leaf size 3, and minimum split size 2 for Sringeri; for Sulya it was 300 trees with unrestricted depth, minimum leaf size 1, and minimum split size 2. The selected SVR used $C = 10$, $\gamma = 0.1$, and $\epsilon = 0.1$ for Sringeri; Sulya used $C = 10$, $\gamma = 0.1$, and $\epsilon = 1.0$.

9 Project artifacts

`outputs/cleaned_rainfall.csv` Calendar-complete observations, one row per month and location; missing targets remain missing.

`outputs/rainfall_features.csv` Modelling features and target.

`outputs/rainfall_monthly_features.csv` Join-ready observed values and features with quality metadata.

`outputs/data_quality_report.json` Coverage, gaps, duplicates, manual-review values, and assumptions.

`outputs/model_metrics.csv` Cross-validation and holdout metrics, separately labelled.

`outputs/holdout_predictions.csv` Observations and predictions for each model and baseline.

`outputs/cross_validation_results.csv` Fold-level validation results, including fold date ranges, for the parameter search.

`outputs/training_report.md` Run summary, best holdout MAE model per location, caveats, and plot references.

`outputs/rainfall_analysis_report.md` Comprehensive source audit, EDA, temporal diagnostics, selected CV folds, holdout, baseline, and interpretation report.

`outputs/eda/` Unit/source audit, observation coverage, annual totals with incomplete years flagged, monthly profiles, and matched-period comparison tables.

`outputs/analysis/` Exact-calendar autocorrelation, lag relationships, prior three/six-month rainfall totals, seasonal totals, and integration-ready monthly observations/features.

`outputs/imd/source_catalogue.csv` Catalogue of WRD-SMS and official IMD products, including unit, period, spatial scale, and actual download status.

`outputs/imd/data_dictionary.md` Field definitions and missing-data, unit, and source-separation rules for the independent IMD ingestion path.

`outputs/imd/` Populated with normalized IMD daily/monthly CSVs, metadata, and separate plots only after a local file is successfully inspected and ingested. No IMD observations were downloaded in this project run.

`docs/imd_data_acquisition_report.md` Official source audit, attempted-download outcomes, local acquisition steps, and caveats.

`outputs/plots/` Historical series, missingness, monthly distributions, residuals, model comparisons, feature importance, and holdout time series.

`models/` Joblib model bundles with feature order, training range, selected parameters, random seed, data fingerprint, and dependency versions.

10 Installation and execution

From the project root, create an environment, install dependencies, run training, and run the automated tests:

```
python -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python scripts/train_rainfall.py
python -m pytest -q
```

The training command does not modify the original workbooks. It regenerates the CSV reports, saved models, and PNG plots. To inspect or ingest separately downloaded IMD files, follow the commands in Section 2 and the acquisition report. The IMD ingestion command does not retrain Random Forest or SVR.

Automated tests

The repository tests check workbook parsing, missing months, duplicate behavior, location separation, feature alignment and leakage, chronological separation, metric behavior, model prediction dimensions, and model reload/inference. The separate IMD tests check date parsing, missing-value preservation, unit validation and conversion, source-type separation, and duplicate rejection. On 20 October 2026, `.venv/bin/python -m pytest -q` executed successfully: 17 tests passed. IMD tests use minimal temporary software fixtures; these are not research observations. `compileall` also completed successfully. The IMD tests validate local parsing behavior; they do not validate any unavailable NWDP file or claim that an IMD download succeeded.

11 Using a saved model for inference

Each model bundle is tied to a location and records the required feature names and order. In Python, load the feature export, select rows for the desired location, and call the documented inference helper:

```
import pandas as pd
from src.inference import predict

features = pd.read_csv("outputs/rainfall_features.csv")
new_rows = features.loc[features["location"].eq("Sringeri")].tail(1)
values = predict(
    "models/sringeri_randomforest_holdout.pkl",
    new_rows,
)
print(values) # predicted rainfall in mm
```

For future months, construct the same feature columns in the saved feature order. Month sine/cosine are based on the target month; rainfall lags and rolling means must use only rainfall observations available before that month. If a prior observation is missing, leave the corresponding feature missing; the fitted pipeline applies its training-derived median imputer. The example above uses the historical feature export to demonstrate invocation and is not a future forecast.

12 Interpretation and limitations

The series are short and contain a nine-month gap in 2023. The 12-point holdout is small, and several holdout rainfall totals are much larger than dry-season values. Metrics are therefore sensitive to a few months. High values are retained and included in the quality report for manual review rather than automatically removed. The unit is verified as mm from the WRD-SMS-labelled Karnataka WRIS source; the monthly completeness convention remains undocumented and the source marks values provisional. A rainfall-only regression model predicts rainfall only; it does not identify invasive species or establish a causal ecological relationship.